



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 2 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Develop the webpage for an online shopping website with using HTML, CSS and HTML5 tags And calculate the bill.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ShopEase</title>
  <link rel="stylesheet" href="Case Study 2.css">
</head>
<body>

  <header>
    <h1>ShopEase</h1>
    <p>Browse our catalog, enter quantities, and generate your bill instantly.</p>
  </header>

  <div class="container">
    <section class="products-section">
      <h2>Product Catalog</h2>
      <div class="product-grid">
        <div class="product-card">
          
          <div class="product-name">Wireless Headphones</div>
          <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Premium
noise-canceling headphones with spatial audio, extra-cushioned earcups, and up to 40 hours of continuous
playback.</div>
          <div class="product-price">Rs. 1499.00</div>
          <div class="form-group" style="margin-top: 10px;">
            <label for="qtyHeadphones">Quantity</label>
            <input type="number" id="qtyHeadphones" value="0" min="0">
          </div>
        </div>

        <div class="product-card">
          
          <div class="product-name">Smart Watch</div>
          <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Smart
wearable tracking heart rate, sleep cycles, steps, and blood oxygen. Features a waterproof design and 1.4-
inch AMOLED display.</div>
          <div class="product-price">Rs. 2499.00</div>
          <div class="form-group" style="margin-top: 10px;">
```

```

        <label for="qtyWatch">Quantity</label>
        <input type="number" id="qtyWatch" value="0" min="0">
    </div>
</div>

<div class="product-card">
    
    <div class="product-name">Mechanical Keyboard</div>
    <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Tactile
blue switches with RGB customizable lighting. Perfect for rapid typing or gaming sessions with key
durability of 50M clicks.</div>
    <div class="product-price">Rs. 1999.00</div>
    <div class="form-group" style="margin-top: 10px;">
        <label for="qtyKeyboard">Quantity</label>
        <input type="number" id="qtyKeyboard" value="0" min="0">
    </div>
</div>

<div class="product-card">
    
    <div class="product-name">Wireless Mouse</div>
    <div style="font-size: 12px; color: gray; margin: 8px 0; min-height:
50px;">Comfortable ergonomic optical mouse with quiet click buttons, selectable DPI speeds, and up to 12
months battery life.</div>
    <div class="product-price">Rs. 699.00</div>
    <div class="form-group" style="margin-top: 10px;">
        <label for="qtyMouse">Quantity</label>
        <input type="number" id="qtyMouse" value="0" min="0">
    </div>
</div>

<div class="product-card">
    
    <div class="product-name">Bluetooth Speaker</div>
    <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Rugged
portable speaker with rich acoustic sound, waterproof casing, dual subwoofers, and integrated
microphone.</div>
    <div class="product-price">Rs. 999.00</div>
    <div class="form-group" style="margin-top: 10px;">
        <label for="qtySpeaker">Quantity</label>
        <input type="number" id="qtySpeaker" value="0" min="0">
    </div>
</div>

<div class="product-card">
    
    <div class="product-name">Wireless Earbuds</div>
    <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Compact
in-ear headphones with touch controls, instant pairing, sweat resistance, and compact magnetic charging
case.</div>
    <div class="product-price">Rs. 1299.00</div>
    <div class="form-group" style="margin-top: 10px;">
        <label for="qtyEarbuds">Quantity</label>
        <input type="number" id="qtyEarbuds" value="0" min="0">
    </div>
</div>

<div class="product-card">
    
    <div class="product-name">Laptop Stand</div>
    <div style="font-size: 12px; color: gray; margin: 8px 0; min-height: 50px;">Adjustable
aluminum stand offering six levels of height adjustment to reduce neck strain. Keeps your computer cool
with ventilation slots.</div>
    <div class="product-price">Rs. 799.00</div>
    <div class="form-group" style="margin-top: 10px;">

```

```

                <label for="qtyStand">Quantity</label>
                <input type="number" id="qtyStand" value="0" min="0">
            </div>
        </div>
    </div>
</section>

<section class="cart-section">
    <h2>Billing Details</h2>

    <div class="form-group">
        <label for="customerName">Customer Name</label>
        <input type="text" id="customerName" placeholder="Enter name">
    </div>

    <div class="form-group">
        <label for="shippingSelect">Shipping Option</label>
        <select id="shippingSelect">
            <option value="0">Standard Shipping (FREE)</option>
            <option value="150">Express Shipping (Rs. 150.00)</option>
        </select>
    </div>

    <div class="form-group">
        <label for="promoCode">Promo Code (WELCOME10, FESTIVE15, SAVE20, SUPER30)</label>
        <input type="text" id="promoCode" placeholder="Enter code">
    </div>

    <button class="btn-checkout" id="billBtn">Generate Bill / Checkout</button>
</section>
<footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px
solid lightgray; padding-top: 15px;">
    Made by Dhanish Ladwani PRN - 24070521020
</footer>
</div>

<div id="receiptModal" class="modal-overlay">
    <div class="modal-content">
        <div class="modal-body">
            <div id="receiptBox" class="receipt-box">
            </div>
        </div>
        <div class="modal-footer">
            <div class="modal-actions">
                <button class="btn-modal btn-close" id="closeBtn">Close</button>
                <button class="btn-modal btn-print" id="printBtn">Print Receipt</button>
            </div>
        </div>
    </div>
</div>

<script src="Case Study 2.js"></script>
</body>
</html>

```

Case Study 2.css

```

body {
    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
    background-color: whitesmoke;
    margin: 0;
    padding: 20px;
    color: black;
    line-height: 1.6;
}

h1, h2, h3 {
    text-align: center;
}

```

```
    color: black;
    margin-bottom: 15px;
    letter-spacing: -1px;
    font-weight: 700;
}

h1 {
    font-size: 32px;
}

h2 {
    font-size: 24px;
    border-bottom: 2px solid lightgray;
    padding-bottom: 8px;
    margin-top: 10px;
}

header {
    background-color: white;
    padding: 25px;
    border: 1px solid lightgray;
    border-radius: 8px;
    margin-bottom: 25px;
    text-align: center;
}

header h1 {
    margin-bottom: 8px;
    font-size: 36px;
}

header p {
    color: gray;
    margin: 0;
    font-size: 16px;
}

.container {
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 8px;
    margin: auto;
    max-width: 960px;
}

.products-section {
    margin-bottom: 30px;
}

.product-grid {
    text-align: center;
}

.product-card {
    display: inline-block;
    width: 29%;
    margin: 1.5%;
    border: 1px solid lightgray;
    border-radius: 6px;
    padding: 15px;
    background-color: white;
    vertical-align: top;
    text-align: center;
    box-sizing: border-box;
    transition: background-color 0.2s, border-color 0.2s;
}

.product-card:hover {
```

```
    background-color: aliceblue;
    border-color: gray;
}

.product-img {
    width: 100%;
    height: 130px;
    object-fit: cover;
    border-radius: 4px;
    background-color: lightgray;
}

.product-name {
    font-weight: bold;
    margin-top: 12px;
    font-size: 18px;
    letter-spacing: -0.5px;
}

.product-price {
    color: royalblue;
    font-weight: bold;
    font-size: 20px;
    margin: 8px 0;
    letter-spacing: -0.5px;
}

.cart-section {
    border-top: 1px solid lightgray;
    padding-top: 25px;
    margin-top: 25px;
}

.form-group {
    margin-bottom: 18px;
    max-width: 500px;
    margin-left: auto;
    margin-right: auto;
}

.form-group label {
    display: block;
    font-weight: bold;
    margin-bottom: 6px;
    color: black;
    font-size: 14px;
}

.form-group input, .form-group select {
    width: 100%;
    padding: 10px;
    border: 1px solid lightgray;
    border-radius: 4px;
    box-sizing: border-box;
    font-size: 15px;
    line-height: 1.4;
}

.form-group input:focus, .form-group select:focus {
    border-color: gray;
    outline: none;
}

.bill-breakdown {
    background-color: whitesmoke;
    border: 1px solid lightgray;
    border-radius: 6px;
    padding: 20px;
    margin: 20px auto;
```

```
    max-width: 500px;
    box-sizing: border-box;
    font-size: 15px;
}

.bill-row {
    margin-bottom: 10px;
}

.bill-row span {
    display: inline-block;
    width: 49%;
}

.bill-row span:last-child {
    text-align: right;
    font-weight: bold;
}

.bill-row.total {
    font-weight: bold;
    border-top: 1px solid lightgray;
    padding-top: 10px;
    margin-top: 10px;
    font-size: 20px;
    letter-spacing: -0.5px;
}

.bill-row.total span:last-child {
    color: forestgreen;
}

.btn-checkout {
    display: block;
    width: 100%;
    max-width: 500px;
    margin: 20px auto 0 auto;
    padding: 12px;
    background-color: forestgreen;
    color: white;
    border: none;
    border-radius: 4px;
    font-weight: bold;
    font-size: 16px;
    cursor: pointer;
    text-align: center;
    transition: background-color 0.2s;
    letter-spacing: -0.5px;
}

.btn-checkout:hover {
    background-color: darkgreen;
}

.modal-overlay {
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background-color: rgba(0,0,0,0.5);
    display: none;
    z-index: 100;
}

.modal-overlay.open {
    display: block;
}
```

```
.modal-content {
  background-color: white;
  margin: 8% auto;
  padding: 25px;
  border: 1px solid lightgray;
  border-radius: 8px;
  width: 85%;
  max-width: 480px;
}

.receipt-box {
  border: 1px dashed black;
  padding: 20px;
  background-color: white;
  font-family: monospace;
  color: black;
  font-size: 14px;
  line-height: 1.5;
}

.receipt-title {
  text-align: center;
  font-weight: bold;
  font-size: 18px;
  margin-bottom: 12px;
  border-bottom: 1px dashed black;
  padding-bottom: 8px;
}

.receipt-row {
  margin-bottom: 6px;
}

.receipt-row span {
  display: inline-block;
  width: 49%;
}

.receipt-row span:last-child {
  text-align: right;
}

.receipt-divider {
  border-bottom: 1px dashed black;
  margin: 12px 0;
}

.modal-actions {
  margin-top: 20px;
  text-align: right;
}

.btn-modal {
  padding: 10px 20px;
  border-radius: 4px;
  font-weight: bold;
  cursor: pointer;
  font-size: 14px;
  letter-spacing: -0.5px;
}

.btn-close {
  background-color: lightgray;
  border: 1px solid lightgray;
  color: black;
}

.btn-close:hover {
  background-color: gray;
}
```

```

}

.btn-print {
  background-color: royalblue;
  border: 1px solid royalblue;
  color: white;
  margin-left: 10px;
}

.btn-print:hover {
  background-color: navy;
}

@media print {
  body > :not(.modal-overlay) {
    display: none;
  }
  .modal-overlay {
    display: block;
    position: absolute;
    left: 0;
    top: 0;
    width: 100%;
    background-color: white;
  }
  .modal-content {
    border: none;
    margin: 0 auto;
    width: 100%;
  }
  .modal-actions {
    display: none;
  }
}

```

Case Study 2.js

```

document.getElementById("billBtn").onclick = () => {
  var customerName = document.getElementById("customerName").value.trim();
  var q1 = Number(document.getElementById("qtyHeadphones").value) || 0;
  var q2 = Number(document.getElementById("qtyWatch").value) || 0;
  var q3 = Number(document.getElementById("qtyKeyboard").value) || 0;
  var q4 = Number(document.getElementById("qtyMouse").value) || 0;
  var q5 = Number(document.getElementById("qtySpeaker").value) || 0;
  var q6 = Number(document.getElementById("qtyEarbuds").value) || 0;
  var q7 = Number(document.getElementById("qtyStand").value) || 0;

  (customerName === "") && alert("Please enter Customer Name!");
  (customerName === "") && document.getElementById("customerName").focus();

  (customerName !== "" && q1 === 0 && q2 === 0 && q3 === 0 && q4 === 0 && q5 === 0 && q6 === 0 && q7 === 0) && alert("Please select at least one item quantity!");

  var isValid = (customerName !== "") && (q1 > 0 || q2 > 0 || q3 > 0 || q4 > 0 || q5 > 0 || q6 > 0 || q7 > 0);

  var subtotal = (q1 * 1499.00) + (q2 * 2499.00) + (q3 * 1999.00) + (q4 * 699.00) + (q5 * 999.00) + (q6 * 1299.00) + (q7 * 799.00);

  var code = document.getElementById("promoCode").value.trim().toUpperCase();
  var discountPercentage = (code === "WELCOME10") ? 10 : ((code === "FESTIVE15") ? 15 : ((code === "SAVE20") ? 20 : ((code === "SUPER30") ? 30 : 0)));

  var discountAmount = subtotal * (discountPercentage / 100);
  var amountAfterDiscount = subtotal - discountAmount;

  var gstAmount = amountAfterDiscount * 0.18;
  var shippingCost = Number(document.getElementById("shippingSelect").value) || 0;

```



```
document.getElementById("closeBtn").onclick = () => {  
    document.getElementById("receiptModal").classList.remove("open");  
};  
  
document.getElementById("printBtn").onclick = () => {  
    window.print();  
};
```

4. Output:

ShopEase

Ask Gemini

C:/Users/Admin/Downloads/Javascript/Case%20Study%202.html

ShopEase

Browse our catalog, enter quantities, and generate your bill instantly.

Product Catalog



Wireless Headphones
Premium noise-canceling headphones with spatial audio, extra-cushioned earcups, and up to 40 hours of continuous playback.
Rs. 1499.00
Quantity
0



Smart Watch
Smart wearable tracking heart rate, sleep cycles, steps, and blood oxygen. Features a waterproof design and 1.4-inch AMOLED display.
Rs. 2499.00
Quantity
0



Mechanical Keyboard
Tactile blue switches with RGB customizable lighting. Perfect for rapid typing or gaming sessions with key durability of 50M clicks.
Rs. 1999.00
Quantity
0



Wireless Mouse
Comfortable ergonomic optical mouse with quiet click buttons, selectable DPI speeds, and up to 12 months battery life.
Rs. 699.00
Quantity
2



Bluetooth Speaker
Rugged portable speaker with rich acoustic sound, waterproof casing, dual subwoofers, and integrated microphone.
Rs. 999.00
Quantity
1



Wireless Earbuds
Compact in-ear headphones with touch controls, instant pairing, sweat resistance, and compact magnetic charging case.
Rs. 1299.00
Quantity
0

Billing Details

Customer Name

Dhanish Ladwani

Shipping Option

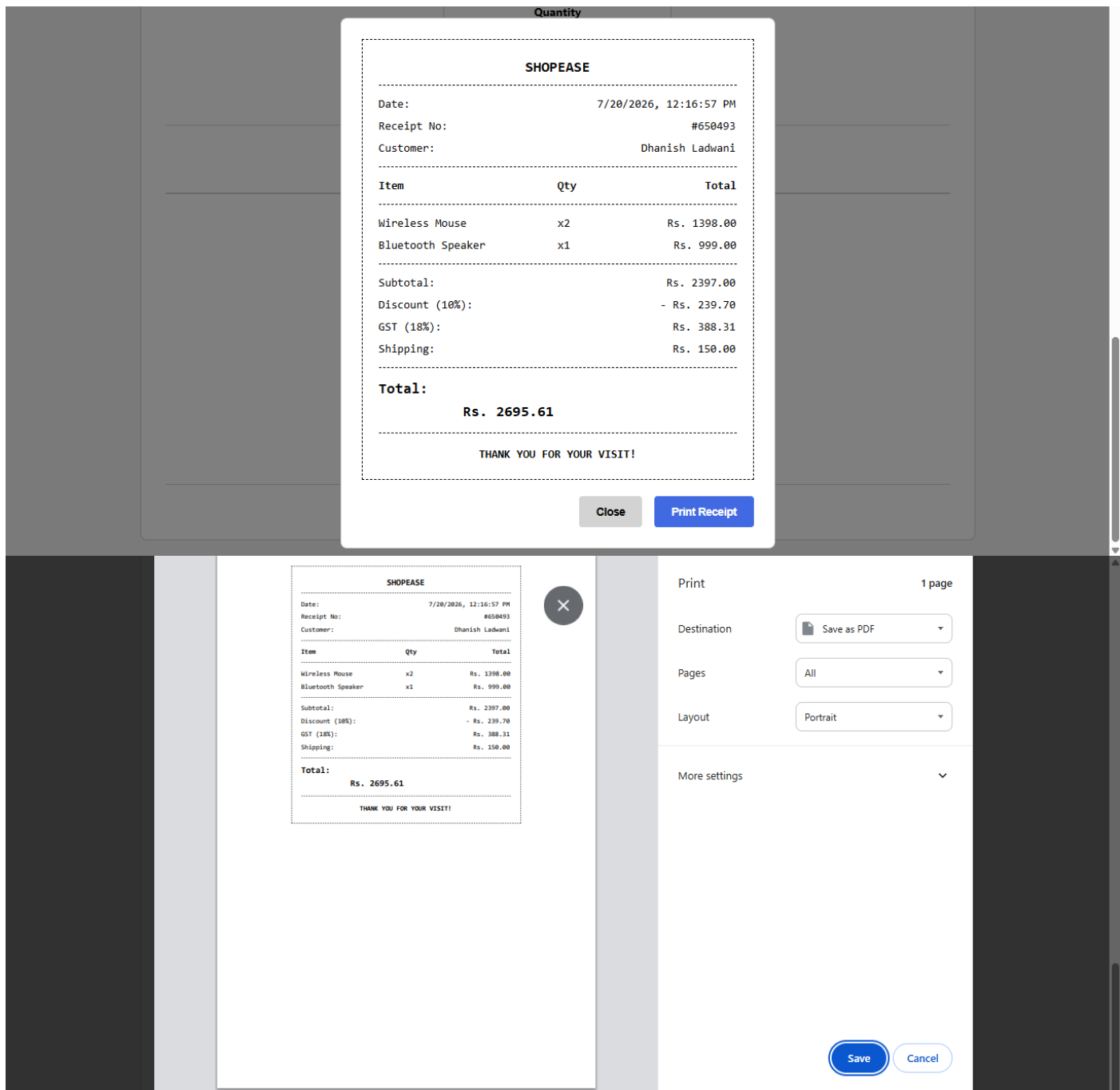
Express Shipping (Rs. 150.00)

Promo Code (WELCOME10, FESTIVE15, SAVE20, SUPER30)

WELCOME10

Generate Bill / Checkout

Made by Dhanish Ladwani PRN - 24070521020



5. Result/Conclusion:

A multi-product e-commerce billing system, ShopEase, was built where customers can select quantities for multiple products from a catalog, apply promo codes for discounts, choose a shipping option, and generate a detailed printable receipt with subtotal, discount, GST (18%) and shipping charges computed dynamically. This case study reinforced DOM manipulation, form handling and calculation logic using JavaScript's Math and Date objects.